

Statistics 6111

Theory and Applications of Linear Models

Summary Notes

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1 Matrix Preliminaries: Generalized Inverses

Theorem 1.1. For any matrix A of rank $r \geq 1$, there exist nonsingular matrices P and Q such that

$$PAQ = \begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix},$$

where D is a nonsingular $r \times r$ diagonal matrix.

The matrix

$$G = Q \begin{bmatrix} D^{-1} & F \\ H & B \end{bmatrix} P$$

satisfies $AGA = A$, where F, H , and B are any matrices of appropriate dimensions.

Definition. A generalized inverse of a matrix A is any matrix G that satisfies $AGA = A$.

Subsequently, for any matrix A , we use A^- to denote an arbitrary generalized inverse of A .

Definition. The equations $A\underline{x} = \underline{y}$ are said to be consistent if they have at least one solution.

Theorem 1.2. For any matrix A , a matrix G is a generalized inverse of A if and only if $\underline{x} = G\underline{y}$ is a solution to the equations $A\underline{x} = \underline{y}$ for every vector $\underline{y}$ for which the equations are consistent.

Remark. Theorem 1.2 can be extended to the matrix equation $AX = Y$. That is, for any matrix A , a matrix G is a generalized inverse of A if and only if $X = GY$ is a solution to the matrix equation $AX = Y$ for every matrix Y for which the equation is consistent.

Theorem 1.3. All solutions to a consistent matrix equation $AX = Y$ are generated by putting

$$X = A^-Y + (I - A^-A)Z$$

and letting Z range over the space of all possible matrices (of appropriate dimensions), where I denotes the identity matrix (of appropriate dimensions).

Lemma 1.1. A matrix G is a generalized inverse of a matrix A if and only if G' is a generalized inverse of A' , where A' is the transpose of A .

Corollary. If A is a symmetric matrix, then $(A^-)'$ is a generalized inverse of A . In particular, for any matrix X , $[(X'X)^-]'$ is a generalized inverse of $X'X$.

Lemma 1.2. For any matrix X , $\text{tr}(X'X) = 0$ implies $X = 0$.

Lemma 1.3. For any matrix X and for any matrices P and Q of appropriate dimensions, we have (i) $PX'X = QX'X$ implies $PX' = QX'$ and (ii) $X'XP = X'XQ$ implies $XP = XQ$.

Lemma 1.4. For any matrix X , $X'X(X'X)^{-}X' = X'$, that is, $X(X'X)^{-}$ is a generalized inverse of X' .

Corollary 1. (i) $X'X[(X'X)^{-}]'X' = X'$, (ii) $X(X'X)^{-}X'X = X$, and (iii) $X[(X'X)^{-}]'X'X = X$.

Corollary 2. Corresponding to any matrix X , there exists a matrix B satisfying $X'XB = X'$.

Lemma 1.5. Let X be any matrix and let B_1 and B_2 be any two matrices that satisfy $X'XB = X'$. Then, $XB_1 = XB_2$.

Corollary 1. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, $XBX = B'X'X = X$.

Corollary 2. Let X be any matrix and let G_1 and G_2 be any two generalized inverses of $X'X$. Then, $XG_1X' = XG_2X'$.

Corollary 3. For any matrix X , $X(X'X)^{-}X' = X[(X'X)^{-}]'X'$, that is, $X(X'X)^{-}X'$ is symmetric.

Corollary 4. Let X be any matrix and let B be any matrix that satisfies $X'XB = X'$. Then, the matrix XB is symmetric.

Let $P_X = XB$ where B is a solution to $X'XB = X'$. Then, P_X has the following properties.

- (1) P_X is invariant to the choice of solution B (by Lemma 1.5).
- (2) P_X is symmetric (by Corollary 4 to Lemma 1.5).
- (3) $P_X = X(X'X)^{-}X'$ (by Lemma 1.4).
- (4) $P_XX = X$ (by Corollary 1 to Lemma 1.5).
- (5) P_X is idempotent, that is, $P_X^2 = P_X$ [by (3) and (4)].

P_X is said to be an orthogonal projection matrix.

Definition. The column space of an $m \times n$ matrix A consists of all $m \times 1$ vectors spanned by the columns of A , that is, a vector $\underline{c}$ belongs to the column space of A if and only if there exists a vector $\underline{b}$ for which $\underline{c} = A\underline{b}$. Similarly, the row space of A consists of all $1 \times n$ vectors spanned by the rows of A .

Notation. We use $\mathcal{C}(A)$ to denote the column space of A and use $\mathcal{R}(A)$ to denote the row space of A .

Lemma 1.6. (i) $\mathcal{C}(B)$ is a subspace of $\mathcal{C}(A)$ if and only if $B = AC$ for some matrix C . (ii) $\mathcal{R}(B)$ is a subspace of $\mathcal{R}(A)$ if and only if $B = CA$ for some matrix C .

Theorem 1.4. Let $M = \begin{bmatrix} A & C \\ D & B \end{bmatrix}$ be a partitioned matrix.

(i) If $\mathcal{C}(C)$ is a subspace of $\mathcal{C}(A)$ and $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(A)$, then the partitioned matrix

$$\begin{bmatrix} A^- + A^-CQ^-DA^- & -A^-CQ^- \\ -Q^-DA^- & Q^- \end{bmatrix}$$

is a generalized inverse of M , where $Q = B - DA^-C$.

(ii) If $\mathcal{C}(D)$ is a subspace of $\mathcal{C}(B)$ and $\mathcal{R}(C)$ is a subspace of $\mathcal{R}(B)$, then the partitioned matrix

$$\begin{bmatrix} P^- & -P^-CB^- \\ -B^-DP^- & B^- + B^-DP^-CB^- \end{bmatrix}$$

is a generalized inverse of M , where $P = A - CB^-D$.

Theorem 1.5. Let $W = R + D$, where $\mathcal{C}(D)$ is a subspace of $\mathcal{C}(R)$ and $\mathcal{R}(D)$ is a subspace of $\mathcal{R}(R)$ (as would be the case, for example, if R were nonsingular). Take S, T , and U to be any three matrices such that

$$D = STU.$$

Then, the matrix

$$R^- - R^-ST(T + TUR^-ST)^-TUR^-$$

is a generalized inverse of W .

2 Classical (Fixed-Effects) Estimation Theory

2.1 Linear Models

Let $\underline{y}$ represent an $n \times 1$ vector of (scalar-valued) data points. We suppose that

$$\underline{y} = X\underline{\beta} + \underline{\epsilon},$$

where X is a given $n \times p$ “design” matrix, $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters that is contained in some given parameter space, and $\underline{\epsilon}$ is an $n \times 1$ vector of unobservable “residuals” or “deviations.”

It is convenient to introduce various special cases of this “linear model.”

Model I (Least squares model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; parameter space = $\{\underline{\beta}: \underline{\beta} \in R^p\}$ (where R^p denotes the Euclidean p -space).

Model II (Gauss-Markov model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = \sigma^2 I$; parameter space = $\{\underline{\beta}, \sigma^2: \underline{\beta} \in R^p, \sigma^2 > 0\}$.

Model III (Aitken model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = \sigma^2 H$, where H is a given positive definite or positive semidefinite matrix; parameter space = $\{\underline{\beta}, \sigma^2: \underline{\beta} \in R^p, \sigma^2 > 0\}$.

Model IV (Mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon} = Z_1 \underline{b}_1 + \cdots + Z_m \underline{b}_m + \underline{e}$, where Z_i is a given $n \times q_i$ “design” matrix, $\underline{b}_i$ is a $q_i \times 1$ random vector such that $E(\underline{b}_i) = \underline{0}$ and $\text{Var}(\underline{b}_i) = \sigma_i^2 I$, and $\underline{e}$ is an $n \times 1$ random vector such that $E(\underline{e}) = \underline{0}$ and $\text{Var}(\underline{e}) = \sigma^2 I$, and where $E(\underline{b}_i \underline{e}') = 0$ and $E(\underline{b}_i \underline{b}_j') = 0$ for $i = 1, \dots, m$, $j = 1, \dots, m$, and $i \neq j$; parameter space = $\{\underline{\beta}, \sigma_1^2, \dots, \sigma_m^2, \sigma^2: \underline{\beta} \in R^p, \sigma_1^2 \geq 0, \dots, \sigma_m^2 \geq 0, \sigma^2 > 0\}$. Here, $\sigma_1^2, \dots, \sigma_m^2$, and σ^2 are called variance components.

Model V (Random-effects model): a special case of the mixed-effects model where $X = \underline{1}$ (a column vector of 1's).

Model VI (General mixed-effects model, with a linear variance structure): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = \theta_1 H_1 + \cdots + \theta_m H_m$, where $H_1, \dots, H_m$ are given symmetric matrices; parameter space = $\{\underline{\beta}, \underline{\theta}: \underline{\beta} \in R^p, \underline{\theta} \in \Omega\}$, where $\underline{\theta} = (\theta_1, \dots, \theta_m)'$ and Ω is the set of $\underline{\theta}$ -vectors for which $\text{Var}(\underline{\epsilon})$ is positive definite or positive semidefinite, or Ω is a given subset of that set.

Model VII (General mixed-effects model): $\underline{y} = X\underline{\beta} + \underline{\epsilon}$; $\underline{\epsilon}$ is a random vector such that $E(\underline{\epsilon}) = \underline{0}$ and $\text{Var}(\underline{\epsilon}) = V$, where the elements of V are functions of some m -dimensional parameter vector

$\underline{\theta}$; parameter space = $\{\underline{\beta}, \underline{\theta}: \underline{\beta} \in R^p, \underline{\theta} \in \Omega\}$, where Ω is the set of $\underline{\theta}$ -vectors for which V is positive definite or positive semidefinite, or Ω is a given subset of that set.

In all of Models I—VII, the space of possible values for $\underline{\beta}$ is the Euclidean p -space. Limiting $\underline{\beta}$ to some subset of R^p gives a constrained version of each model. The kinds of limitations on $\underline{\beta}$ that are most commonly imposed are those determined by linear equality constraints or linear inequality constraints, such as $A\underline{\beta} = \underline{b}$ or $A\underline{\beta} \geq \underline{b}$.

Notation. Let $p^* = \text{rank}(X)$.

When $p^* = p$, the linear model is customarily referred to as a full-rank linear model.

2.2 Estimability

Definition. An estimator $t(\underline{y})$ of a linear function $\underline{\lambda}'\underline{\beta}$ is said to be linear if $t(\underline{y}) \equiv c + \underline{a}'\underline{y}$ for some constant c and some vector $\underline{a}$ of constants.

Definition. Under Models II—VII, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] \equiv \underline{\lambda}'\underline{\beta}$.

Lemma 2.1. *Under Models II—VII, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if $c = 0$ and $\underline{\lambda}' = \underline{a}'X$.*

Definition. Under Models II—VII, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Otherwise, it is said to be nonestimable.

Theorem 2.1. *Under Models II—VII, a necessary and sufficient condition for a linear function $\underline{\lambda}'\underline{\beta}$ to be estimable is that its coefficient vector $\underline{\lambda}'$ be expressible in the form $\underline{\lambda}' = \underline{a}'X$, that is, that $\underline{\lambda}'$ belong to $\mathcal{R}(X)$, that is, that $\underline{\lambda}'\underline{\beta}$ be expressible as $\underline{a}'X\underline{\beta}$, that is, that*

$$\text{rank}(X) = \text{rank} \begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix},$$

where $\begin{bmatrix} X \\ \underline{\lambda}' \end{bmatrix}$ is the matrix X augmented by the row vector $\underline{\lambda}'$.

Corollary 1. Under Models II—VII, the elements of the vector $X\underline{\beta}$ are estimable.

Corollary 2. Under Models II—VII, if $\underline{\lambda}'_1\underline{\beta}, \dots, \underline{\lambda}'_r\underline{\beta}$ are estimable, then any linear combination of them is estimable.

Corollary 3. If $p^* = p$, then for every $p \times 1$ vector $\underline{\lambda}$, $\underline{\lambda}'\underline{\beta}$ is estimable.

Definition. Linear functions $\underline{\lambda}'_1 \underline{\beta}, \dots, \underline{\lambda}'_r \underline{\beta}$ are said to be linearly independent if $\underline{\lambda}_1, \dots, \underline{\lambda}_r$ form a linearly independent set of vectors or, equivalently, if the matrix $\Lambda = (\underline{\lambda}_1, \dots, \underline{\lambda}_r)$ has rank r .

Theorem 2.2. *Under Models II—VII, there exists a set of p^* linearly independent functions $\underline{\lambda}'_1 \underline{\beta}, \dots, \underline{\lambda}'_{p^*} \underline{\beta}$ that are estimable. No collection of estimable linear functions can contain more than p^* linearly independent functions.*

Lemma 2.2. *For any matrices A and B (of appropriate dimensions), $\text{rank}(AB) \leq \text{rank}(A)$ and $\text{rank}(AB) \leq \text{rank}(B)$.*

Lemma 2.3. *For any matrix A , $\text{rank}(A' A) = \text{rank}(A)$.*

2.3 Least Squares Estimation

Definition. Under Models II—VII, a (simple or ordinary) least squares estimate of an estimable function $\underline{\lambda}' \underline{\beta}$ is $\underline{\lambda}' \hat{\underline{\beta}}$, where $\hat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes the sum of squares

$$Q(\underline{\beta}) = (\underline{y} - X\underline{\beta})'(\underline{y} - X\underline{\beta}).$$

A necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \hat{\underline{\beta}}$ is that $\hat{\underline{\beta}}$ satisfy the normal equations

$$X' X \hat{\underline{\beta}} = X' \underline{y}.$$

Theorem 2.3. *The normal equations are consistent.*

Theorem 2.4. *$Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \hat{\underline{\beta}}$ if and only if $\hat{\underline{\beta}}$ is a solution to the normal equations.*

Theorem 2.5. *Under Models II—VII, $\underline{\lambda}' \hat{\underline{\beta}} = \underline{\lambda}' \tilde{\underline{\beta}}$ for any two solutions $\hat{\underline{\beta}}$ and $\tilde{\underline{\beta}}$ to the normal equations if and only if $\underline{\lambda}' \underline{\beta}$ is estimable.*

Corollary. $X \hat{\underline{\beta}} = X \tilde{\underline{\beta}}$ for any two solutions $\hat{\underline{\beta}}$ and $\tilde{\underline{\beta}}$ to the normal equations.

2.4 Geometric Considerations

Definition. Two $t \times 1$ vectors $\underline{a}$ and $\underline{b}$ are said to be orthogonal if their inner product $\underline{a}' \underline{b}$ equals zero.

Definition. Let l_1 be the directed line segment from the origin $\underline{0}$ to a point $\underline{a} \neq \underline{0}$, and let l_2 be the directed line segment from $\underline{0}$ to a point $\underline{b} \neq \underline{0}$. The angle θ formed by l_1 and l_2 is defined by

$$\cos \theta = \frac{\underline{a}' \underline{b}}{[\underline{a}' \underline{a}]^{1/2} [\underline{b}' \underline{b}]^{1/2}}, \quad 0 \leq \theta \leq \pi.$$

When $\theta = \pi/2$, that is, when $\cos \theta = 0$, that is, when $\underline{a}$ and $\underline{b}$ are orthogonal, the directed line segments (or the lines defined by the directed line segments) are said to be perpendicular.

Notation. Let $\mathcal{C}^\perp(X)$ represent the null space of X' , that is, the set of all column vectors that are orthogonal to every column of X [or, equivalently, to every vector in $\mathcal{C}(X)$]. Similarly, $\mathcal{R}^\perp(X)$ denotes the set of all row vectors that are orthogonal to every row of X .

Let $\underline{\hat{y}} = X\underline{\hat{\beta}}$ and $\underline{\hat{\epsilon}} = \underline{y} - \underline{\hat{y}} = \underline{y} - X\underline{\hat{\beta}}$, where $\underline{\hat{\beta}}$ is any solution to the normal equations.

Theorem 2.6. Any $n \times 1$ vector $\underline{y}$ admits a unique decomposition satisfying $\underline{y} = \underline{y}_1 + \underline{y}_2$, where $\underline{y}_1 \in \mathcal{C}(X)$ and $\underline{y}_2 \in \mathcal{C}^\perp(X)$. Moreover, this decomposition is given by $\underline{y}_1 = \underline{\hat{y}} = X\underline{\hat{\beta}}$ and $\underline{y}_2 = \underline{\hat{\epsilon}} = \underline{y} - \underline{\hat{y}} = \underline{y} - X\underline{\hat{\beta}}$, where $\underline{\hat{\beta}}$ is any solution to the normal equations.

Corollary. If W is any matrix with its column space identical to that of the matrix X [that is, $\mathcal{C}(W) = \mathcal{C}(X)$], then $W\underline{\hat{\tau}} = X\underline{\hat{\beta}}$, where $\underline{\hat{\beta}}$ is any solution to $X'X\underline{\hat{\beta}} = X'\underline{y}$ and $\underline{\hat{\tau}}$ is any solution to $W'W\underline{\hat{\tau}} = W'\underline{y}$.

Note: $\underline{\hat{\beta}} \equiv (X'X)^-X'\underline{y}$ solves $X'X\underline{\hat{\beta}} = X'\underline{y}$, implying that $\underline{\hat{y}} = X\underline{\hat{\beta}} = P_X\underline{y}$ and $\underline{\hat{\epsilon}} = \underline{y} - \underline{\hat{y}} = (I - P_X)\underline{y}$

2.5 The Conjugate Normal Equations

Theorem 2.7. Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if the conjugate normal equations

$$X'X\underline{r} = \underline{\lambda}$$

are consistent, that is, if and only if there exists an $\underline{r}$ such that $X'X\underline{r} = \underline{\lambda}$.

Corollary. Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X(X'X)^-\underline{\lambda} = \underline{\lambda}$.

Theorem 2.8. Under Models II—VII, the least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is equal to $\underline{r}'X'\underline{y}$, where $\underline{r}$ is any solution to $X'X\underline{r} = \underline{\lambda}$.

2.6 Expectations/Variances/Covariances of Least Squares Estimators

For two random vectors $\underline{x} = (x_1, \dots, x_m)'$ and $\underline{z} = (z_1, \dots, z_n)'$, we write $\text{cov}(\underline{x}, \underline{z})$ for the $m \times n$ matrix whose (i, j) th element is the covariance between x_i and z_j . The matrix $\text{Var}(\underline{x}) = \text{cov}(\underline{x}, \underline{x})$ is called the variance-covariance matrix, or simply the covariance matrix, of $\underline{x}$.

Lemma 2.4. If $\underline{x}$ is a random vector and L and M are matrices of constants having the same number of rows as $\underline{x}$ has elements, then

$$\text{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M, \text{ where } \Sigma = \text{Var}(\underline{x}).$$

Corollary. $\text{Var}(L'\underline{x}) = L'\Sigma L$, where $\Sigma = \text{Var}(\underline{x})$.

Definition. For a matrix of constants Λ having p rows, the vector $\Lambda'\underline{\beta}$ is said to be estimable if each of its elements is estimable.

Suppose that $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ (where Λ_1 and Λ_2 are matrices of constants having p rows each) are both estimable. Let $\widehat{\underline{\beta}}$ represent any solution to the normal equations, and take R_1 and R_2 to be any two matrices satisfying

$$X'XR_1 = \Lambda_1 \quad \text{and} \quad X'XR_2 = \Lambda_2.$$

The least squares estimators of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, that is, the vectors whose elements are the ordinary least squares estimators of the corresponding elements of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, are given by

$$\Lambda_1'\widehat{\underline{\beta}} = R_1'X'y \quad \text{and} \quad \Lambda_2'\widehat{\underline{\beta}} = R_2'X'y.$$

Under Model VII,

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \Lambda_1'(X'X)^{-1}X'VX(X'X)^{-1}\Lambda_2,$$

and, in particular,

$$\text{Var}(\Lambda_1'\widehat{\underline{\beta}}) = \Lambda_1'(X'X)^{-1}X'VX(X'X)^{-1}\Lambda_1.$$

Under Model II,

$$\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(X'X)^{-1}\Lambda_2,$$

and, in particular,

$$\text{Var}(\Lambda_1'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(X'X)^{-1}\Lambda_1.$$

2.7 The Gauss-Markov Theorem

Theorem 2.9. (The Gauss-Markov Theorem) *Under Models II—VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under Model II, the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the best (minimum variance) linear unbiased estimator (BLUE) of $\underline{\lambda}'\underline{\beta}$.*

Definitions. If $\underline{x}$ is a column vector of r real variables and A is an $r \times r$ matrix of constants, then we refer to the function

$$q(\underline{x}) = \underline{x}'A\underline{x}$$

as a quadratic form. The quadratic form $q(\underline{x})$ is said to be: positive definite if $q(\underline{x}) > 0$ for all nonzero real vectors $\underline{x}$; nonnegative definite if $q(\underline{x}) \geq 0$ all real vectors $\underline{x}$; and positive semidefinite if $q(\underline{x})$ is nonnegative definite but not positive definite (i.e., $q(\underline{x}) = 0$ holds for some nonzero $\underline{x}$).

A square matrix A is said to be positive definite, nonnegative definite, or positive semidefinite, if A is symmetric and the associated quadratic form $\underline{x}'A\underline{x}$ is positive definite, nonnegative definite, or positive semidefinite, respectively.

Theorem 2.10. *Under Model II, suppose that $\Lambda'\underline{\beta}$ is any k -dimensional estimable vector and that $\underline{c} + A'\underline{y}$ is any vector of linear unbiased estimators of the elements of $\Lambda'\underline{\beta}$. Then, the matrix $\text{Var}(\underline{c} + A'\underline{y}) - \text{Var}(\Lambda'\widehat{\underline{\beta}})$ is nonnegative definite, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.*

Corollary 1. Under the condition of Theorem 2.10,

$$\text{tr}[\text{Var}(\underline{c} + A'\underline{y})] \geq \text{tr}[\text{Var}(\Lambda'\widehat{\underline{\beta}})].$$

Corollary 2. Under the condition of Theorem 2.10,

$$\det[\text{Var}(\underline{c} + A'\underline{y})] \geq \det[\text{Var}(\Lambda'\widehat{\underline{\beta}})].$$

Definition. The determinant of the variance-covariance matrix of an estimator of a vector of parametric functions is referred to as the generalized variance of the estimator.

2.8 Estimation of the Residual Variance

We now consider the estimation of σ^2 for Model II.

Definition. A square matrix A is said to be an idempotent matrix if $A^2 = A$.

Lemma 2.5. *For any square matrix A such that $A^2 = mA$ for some scalar m ,*

$$\text{tr}(A) = m \text{rank}(A).$$

Corollary. If A is idempotent, then $\text{tr}(A) = \text{rank}(A)$.

Lemma 2.6. *For any matrix A , P_A and $I - P_A$ are both symmetric idempotent matrices, where $P_A = A(A'A)^{-}A'$.*

Lemma 2.7. *For any $n \times p$ matrix A , $\text{rank}(P_A) = \text{rank}(A)$ and $\text{rank}(I - P_A) = n - \text{rank}(A)$.*

Lemma 2.8. *If $\underline{x}$ is an $n \times 1$ random vector having mean vector $\underline{\mu}$ and variance-covariance matrix Σ and if A is an $n \times n$ matrix of constants, then*

$$E(\underline{x}'A\underline{x}) = \text{tr}(A\Sigma) + \underline{\mu}'A\underline{\mu}.$$

Let $\widehat{\underline{\beta}}$ be any solution to the normal equations. Under Model II,

$$\widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*) = \underline{y}'(\underline{y} - X\widehat{\underline{\beta}})/(n - p^*)$$

is an unbiased estimator of σ^2 .

2.9 Constrained Linear Models

Consider the case where $\underline{\beta}$ is restricted to the subspace of R^p consisting of $\underline{\beta}$ values that satisfy

$$A\underline{\beta} = \underline{b},$$

where A is a given $q \times p$ matrix of rank $q^* \geq 1$ and $\underline{b}$ is a given $q \times 1$ vector. To avoid trivialities, we assume that the linear system $A\underline{\beta} = \underline{b}$ is consistent. We will refer to those modified versions of Models I—VII in which $\underline{\beta}$ is restricted to this subspace as Models Ia—VIIa.

Definition. Under Models IIa—VIIa, an estimator $t(\underline{y})$ is said to be an unbiased estimator of a linear function $\underline{\lambda}'\underline{\beta}$ if $E[t(\underline{y})] = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Lemma 2.9. *Under Models IIa—VIIa, a linear estimator $c + \underline{a}'\underline{y}$ of a linear function $\underline{\lambda}'\underline{\beta}$ is unbiased if and only if there exists a vector $\underline{k}$ such that*

$$c = \underline{k}'\underline{b} \quad \text{and} \quad \underline{\lambda}' = \underline{a}'X + \underline{k}'A.$$

Definition. Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is said to be estimable if there exists a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, if there exists a linear estimator $c + \underline{a}'\underline{y}$ such that $E(c + \underline{a}'\underline{y}) = \underline{\lambda}'\underline{\beta}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

Theorem 2.11. *Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R} \begin{bmatrix} X \\ A \end{bmatrix}$.*

Definition. Under Models IIa—VIIa, a least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ is $\underline{\lambda}'\widehat{\underline{\beta}}$, where $\widehat{\underline{\beta}}$ is any value of $\underline{\beta}$ that minimizes $Q(\underline{\beta})$ for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

A necessary condition for $Q(\underline{\beta})$ to assume a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, is that the constrained normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \widehat{\underline{\beta}} \\ \underline{\rho} \end{bmatrix} = \begin{bmatrix} X'\underline{y} \\ \underline{b} \end{bmatrix}$$

have a solution whose “first” component is $\widehat{\underline{\beta}}$.

Lemma 2.10. *The matrix equation $BW = D$ is consistent, that is, has a solution for W , if and only if $\mathcal{C}(D) \subset \mathcal{C}(B)$ if and only if $\mathcal{C}^\perp(B) \subset \mathcal{C}^\perp(D)$ that is, if and only if $\underline{a}'D = \underline{0}'$ for every vector $\underline{a}$ such that $\underline{a}'B = \underline{0}'$.*

Theorem 2.12. *The constrained normal equations are consistent.*

Theorem 2.13. $Q(\underline{\beta})$ assumes a minimum at $\underline{\beta} = \widehat{\underline{\beta}}$, for $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$, if and only if the constrained normal equations have a solution whose first component is $\widehat{\underline{\beta}}$.

Lemma 2.11.

$$\text{rank} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \text{rank} \begin{bmatrix} X'X \\ A \end{bmatrix} + \text{rank}(A).$$

Lemma 2.12. (i) $\mathcal{R} \begin{bmatrix} X'X \\ A \end{bmatrix}$ is the same as $\mathcal{R} \begin{bmatrix} X \\ A \end{bmatrix}$. (ii) $\text{rank} \begin{bmatrix} X'X \\ A \end{bmatrix} = \text{rank} \begin{bmatrix} X \\ A \end{bmatrix}$.

Corollary to Lemma 2.11.

$$\text{rank} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} = \text{rank} \begin{bmatrix} X \\ A \end{bmatrix} + \text{rank}(A).$$

Theorem 2.14. Let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ represent any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$. Then,

- (i) $A'G_{12}A' = A'G_{21}A' = A'$;
- (ii) $-A'G_{22}A = -A'G_{22}A = X'XG_{21}'A = A'G_{21}X'X = X'XG_{12}A = A'G_{12}'X'X$;
- (iii) $AG_{11}A'$, $AG_{11}X'X$, and $X'XG_{11}A'$ are all zero matrices;
- (iv) $X'XG_{11}X'XG_{11}X'X = X'XG_{11}X'X = X'XG_{11}'X'XG_{11}X'X = X'XG_{11}'X'X$;
- (v) $\text{tr}(X'XG_{11}) = \text{rank} \begin{bmatrix} X'X \\ A \end{bmatrix} - \text{rank}(A) = \text{tr}(X'XG_{11}')$;
- (vi) $X'XG_{11}X'X$ and $A'G_{22}A$ are invariant for any choice of G_{11} and G_{22} .

Lemma 2.13. For any generalized inverse

$$G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \text{ of } \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix},$$

$$X'X = X'XG_{11}X'X - A'G_{22}A.$$

Theorem 2.15. Under Models IIa—VIIa, $\underline{\lambda}'\widehat{\underline{\beta}} = \underline{\lambda}'\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations if and only if $\underline{\lambda}'\underline{\beta}$ is estimable.

Corollary. $X\widehat{\underline{\beta}} = X\widetilde{\underline{\beta}}$ for the first components $\widehat{\underline{\beta}}$ and $\widetilde{\underline{\beta}}$ of any two solutions to the constrained normal equations.

Lemma 2.14. *Under Models IIa—VIIa, a linear function $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $\underline{\lambda}' \in \mathcal{R} \begin{bmatrix} X'X \\ A \end{bmatrix}$.*

The equations that play the role of the conjugate normal equations in our development are the constrained conjugate normal equations

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} \underline{r}_1 \\ \underline{r}_2 \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$

Theorem 2.16. *Under Models IIa—VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable, that is,*

$$\underline{\lambda}' \in \mathcal{R} \begin{bmatrix} X \\ A \end{bmatrix}, \quad \text{that is, } \underline{\lambda}' \in \mathcal{R} \begin{bmatrix} X'X \\ A \end{bmatrix},$$

if and only if the constrained conjugate normal equations are consistent.

Corollary. Under Models IIa—VIIa, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if

$$\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix} \begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}^- \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix} = \begin{bmatrix} \underline{\lambda} \\ \underline{0} \end{bmatrix}.$$